

Athlete Intelligence — Technical Design: Accuracy-First Architecture

Document type: Technical Design Specification
Status: Awaiting approval — no code has been changed
Purpose: Redesign of the intelligence engine to support permanent athlete identities, verified evidence, and continuous monitoring from high school through international level

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1. Design Principles

Before the technical detail, nine principles govern every decision in this design:

#	Principle	Meaning
P1	Evidence-first	No fact is stored without a source. No AI claim is made without a cited DB row.
P2	Append-only	History is never overwritten. Every change creates a new record.
P3	Source hierarchy	Higher-authority sources always win, explicitly, with a documented reason.
P4	Identity permanence	One athlete = one identity forever, from first known result to retirement.
P5	Explicit uncertainty	Estimated values are labelled as estimates. Unknown values are null, not guessed.
P6	Verified before stored	AI extraction is a hypothesis. It is only promoted to verified when cross-checked against a second source or a human confirms.
P7	Auditability	Every stored fact must be traceable to: who found it, when, from where, and with what confidence.
P8	Graceful degradation	If a data source is unavailable, the system stores nothing rather than a guess.
P9	Separation of concerns	Collection, verification, storage, and presentation are separate layers. A bug in presentation must

```
custom_federation_ids JSONB, -- { "federation_code": "athlete_id", ... }

-- Disambiguation fingerprint (computed, used for dedup check)
identity_fingerprint TEXT GENERATED ALWAYS AS (
    lower(canonical_name) || '|' || canonical_nationality || '|' || canonical_sport
) STORED,

-- Status
status TEXT NOT NULL DEFAULT 'active', -- 'active' | 'retired' | 'merged' |
'deceased'
merged_into_id UUID REFERENCES athlete_identities(id), -- if this was a duplicate
identity_confidence INTEGER NOT NULL DEFAULT 70, -- 0-100: how confident we are this is a
unique person
);
```

2. Identity Engine

```
-- Human verification
verified_by TEXT, -- clerk user_id who confirmed identity
verified_at TIMESTAMPTZ
```

2.1 The Problem

CREATE UNIQUE INDEX idx_identity_fingerprint ON athlete_identities(identity_fingerprint);
CREATE INDEX idx_identity_name ON athlete_identities(lower(canonical_name));
A person is not an identity. "James Wilson" might be a high school javelin thrower in Christchurch and an Olympic triathlete from Australia simultaneously. "Brook Macdonald" is unique today, but once the platform covers 100,000 athletes, collisions will occur. The current implementation stores a single name string and relies on Perplexity to "find the right person" — which it demonstrably fails to do.

2.2 Athlete Identity Record

Fingerprint uniqueness, name|nationality|sport as a unique index prevents duplicate creation of the same person. If user submits a "Brook Macdonald" from North Ryecroft, New Zealand

Federations are the common standard. A World Athletics competitor ID uniquely identifies a person in that sport globally. Once confirmed, there's one existing identity, no need for multiple. The system should populate these during identity creation or verification.

Birth year, not full DOB as we encounter athletes who have never disclosed their full date of birth (particularly school-age athletes). Birth year alone is sufficient to distinguish two "James Wilson" athletes in the same sport and nationality.

Merged identities if a duplicate identity is created for the same person (before the dedup check was in place, or because the merged record entered the data feed, confidence as a foreview).
Nonar basis: pending identity hypothesis is merged into the canonical one. The merged record is not deleted — it points to merged_into_id. All child records (evidence, timeline, social) are re-pointed to the canonical identity.

2.4 Handling Athletes with Identical Names

Phase 2 — Confirmation (can be automated or human)
Three-layer disambiguation:

Layer 1: Fingerprint. name|nationality|sport uniqueness means two athletes with the same name but different sports/nationalities are stored as separate identities correctly.

Layer 2: Birth year. If name, nationality, and sport all match, birth year differentiates them (e.g., two New Zealand high jumpers named James Wilson born in 2002 and 2008).

Layer 3: Federation ID. Once a federation ID is attached, it overrides all other disambiguation. Two athletes cannot share a World Athletics ID.

Collision UI. When a user adds an athlete and the fingerprint matches an existing identity, the UI shows: "We already have [Name], [Sport], [Nationality]. Is this the same person?" with options: "Yes, use existing" | "No, this is a different athlete" (which triggers birth year capture).

3. Source Engine

3.1 Source Hierarchy

Sources are ranked in seven tiers. A higher-tier source always supersedes a lower-tier source for the same fact, unless the higher-tier source is more than 12 months older.

4.1 Core Concept

Every fact stored in the system is an evidence-backed claim. There is no "profile data" that can be silently overwritten. Instead, there is a log of claims, each pointing to a source, each with a confidence score and a retrieval timestamp. The currently-believed value for any field is derived by querying which evidence claim is highest-tier, most recent, and uncontradicted.

4.2 Schema

``evidence_items`` table

This is the central table of the entire redesigned system.

70–84	Verified news outlet, named journalist	BBC Sport article with event citation
55–69	AI extraction from cited Perplexity source	Perplexity found the page, GPT extracted the value, URL verified
40–54	AI extraction, source cited but unverifiable	Perplexity cited a URL that returns 404
20–39	AI estimated, no specific source	GPT training knowledge
0–19	Contradicted or speculative	Disputed by higher-authority source

5. Timeline Engine

5.1 Design Concept

A profile is a snapshot. A timeline is the truth. Every meaningful event in an athlete's career is an immutable event on their timeline. Nothing is ever deleted or overwritten — events are added, and corrections create new events that reference and supersede old ones.

5.2 Schema

``timeline_events`` (redesigned)

National rankings	rankings_check	Weekly	Quarterly
Social metrics	social_snapshot	Daily	Weekly
News mentions	news_scan	Daily	Weekly
Sponsorships	sponsor_check	Weekly	Monthly
Team affiliation	team_change	Monthly	Quarterly
Photo	photo_refresh	Monthly	Quarterly
Education	education_check	Monthly	N/A
Injury/return	injury_check	Weekly	N/A
Retirement / return	retirement_check	Weekly	N/A

6.4 What Is Detected and How

Signal	How detected	Source used
New competition results	Perplexity search "athlete_name results site:federation.org"	Tier 1–4 sources
Ranking changes	Federation API or weekly ranking page scrape	Tier 1 API preferred
Sponsorship changes	Perplexity news search + athlete social posts	Tier 6–7
Team changes	Official team announcement pages	Tier 3
Social growth	Platform APIs (Twitter); Perplexity for Instagram/ TikTok estimate	See §7
News mentions	Perplexity news search	Tier 6
Awards	Federation announcements + news	Tier 1–2 + Tier 6
Education	Manual entry or Perplexity + school sports results	Tier 5
Publicly visible milestones	Perplexity + Wikipedia edits	Tier 6–7
Injury/return	News scan (Perplexity)	Tier 6
Retirement / return	Official federation + news	Tier 1 + Tier 6

7. Social Media Engine

7.1 What Is Actually Possible (Compliance Reality)

Platform	Official public API	What's available	Rate limits
X (Twitter)	' v2 API	Follower count, following, tweet count, bio	Free: 500K reads/month
YouTube	' Data API v3	Subscriber count, video count, view count	10,000 units/day
Facebook	'L Public page data removed	Page follower count only with Page access	Not available public

Instagram

'L No public follower count API

Nothing without the user's own
token

Every follower count shown in the UI must have a clear label:

Follower Count	Label
10	is primary
100	BOOLEAN NOT NULL DEFAULT false
1000	-- primary account for this platform
10000	Nothing useful for athletes
100000	Not available public

Data source	UI label	Indicator
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' green checkmark

Do not use your training knowledge! Do not guess.

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2.2 Evidence Ground

Every AI extraction call must follow this contract:

Athlete identity discovery	gpt-4o	Needs broad knowledge to identify sport/nationality
Identity confirmation	gpt-4o-mini	Simple yes/no + brief explanation
Structured data extraction from source text	gpt-4o	High accuracy JSON generation, large context
Social handle extraction	gpt-4o-mini	Simple pattern matching
Photo URL extraction	gpt-4o-mini	Simple URL extraction from short text
Dossier summary generation	gpt-4o	Quality writing, grounded generation
Chat responses	gpt-4o	Tool-calling, multi-step reasoning
Web research (live search)	perplexity/sonar-pro	Only sonar-pro for all paths (not sonar)

All models are accessed via the Replit AI Integrations proxy. No model aliases — only documented model strings.

9. Gap Analysis

9.1 What Exists Now vs What the Vision Requires

Capability	Current State	Vision State	Gap
Athlete identity	athletes.name text column	UUID identity with federation IDs, fingerprint, disambiguation	Complete rebuild
Source tracking	source_domain text per item	sources registry with tier, evidence_items with full provenance	Complete rebuild
Evidence model	Flat columns on athletes table; overwritten on repopulate	Append-only evidence_items with confidence, status, verification	Complete rebuild
Timeline	timeline_events table (partial)	Redesigned with event types, significance, evidence FK, correction model	Major extension
Continuous monitoring	Manual repopulate only	Database job queue, scheduled crawls, change detection	New build
Social data	Flat columns (instagram_followers) overwritten	social_handles + social_snapshots with verification status	Major extension
Social verification	None	3-step handle verification	New build
Source hierarchy	None	7-tier registry with override rules	New build
AI grounding	AI invents from training data	Evidence-only extraction with verbatim cite validation	Complete rebuild
Identity disambiguation	None	Fingerprint index, birth year, federation IDs, collision UI	New build
High school/school athletes	Not considered	education event types, school sports federation sources	Schema extension
Historical data	Overwritten on repopulate	Append-only, superseded not deleted	Requires migration
Conflict resolution	Not considered	disputed status, curator tools	New build

Job queue

Detached promise, lost on restart

PostgreSQL-backed crawl_jobs
table

9.2 Components That Can Be Retained

Component	Retain?	Notes
Clerk authentication	' Fully retained	No change
Stripe billing	' Fully retained	No change
React frontend shell	' Largely retained	New pages for evidence/timeline detail needed
Perplexity integration	' Retained	Upgrade to sonar-pro everywhere; change role to research only
OpenAI integration	' Retained	Model strings corrected; prompts redesigned
Admin panel	' Extended	New job queue monitoring; evidence review tools
contact_enquiries	' Retained	Unaffected
conversations/messages	' Retained (and activated)	Wire up to chat persistence

9.3 Components That Must Be Replaced

Component	Replacement	
athletes table (as a profile store)	athlete_identities (identity) + evidence_items (facts)	
intelligence_items	Folded into evidence_items with `fact_type = 'media_**`	'competition_**'
timeline_events (current)	Redesigned timeline_events with evidence FK	
contacts	Folded into evidence_items with `fact_type = 'coach'	'agent'
competitions	Folded into timeline_events with event_type = 'competition_result'	
auto-populate.ts (current)	New extraction pipeline per §8.2	
photo-lookup.ts (current)	New photo evidence workflow	
Social columns on athletes	social_handles + social_snapshots	

9.4 Estimated Effort

Phase	Description	Estimated effort
Phase 1 — Schema design	Write and review all migrations; Drizzle schema files	1–2 weeks

Discovery, fingerprint, disambiguation,
confirmation flow

		1 week
Phase 3 — Evidence engine	New extraction pipeline, validation, confidence scoring	2 weeks
Phase 4 — Source registry	Populate sources table; tier rules; override logic	3–4 days
Phase 5 — Timeline rebuild	New schema, correction model, evidence FK	1 week
Phase 6 — Social engine	Handle verification, snapshot collection, display labels	1 week
Phase 7 — Job queue	crawl_jobs table, worker loop, scheduler	1 week
Phase 8 — AI layer redesign	Grounded prompts, verbatim validation, chat	1–2 weeks
Week 1–2:	New schema tables created alongside existing tables. Existing system continues to run unchanged.	
Phase 9 — Frontend updates	Evidence viewer, timeline redesign, verification	2 weeks
Week 3–4:	Identity engine deployed. New athletes use new schema. Existing athletes remain in old schema temporarily.	
Week 5–8:	Evidence engine, source registry, new extraction pipeline deployed.	
Phase 10 — Data migration	Port existing athlete data to new schema.	3–4 days
Week 9–10:	Social engine, job queue deployed.	
Total	All monitoring switches to job queue.	~12–14 weeks (one engineer)
Week 11–12:	Frontend updated to read from new schema. Old schema deprecated (kept read-only for 30 days).	
Week 13–14:	Old schema tables dropped. Migration complete.	

10. Migration Path

This approach means the platform remains fully functional throughout the rebuild. Users see improved data quality progressively rather than a downtime event.

Rather than a big-bang rewrite, migration can be done in parallel with the running system using a strangler-fig pattern. See technical design document. No code has been changed. Awaiting approval before implementation begins.

